

Beyond the Rate: Information Content of FOMC Statement Language

Dechang Yu¹ Eileen Zhang²

¹Academy of AI, Xi'an Jiaotong-Liverpool University

²[Affiliation TBC]

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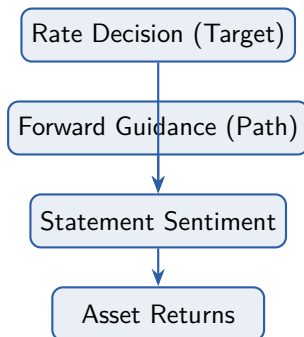
The Puzzle

Central question:

Does FOMC statement language convey information beyond the rate decision?

Why it matters:

- Central banks increasingly rely on forward guidance as a policy tool
- Markets respond to *what the Fed says*, not just what it does
- The “information channel” (Romer & Romer, 2000) suggests FOMC statements reveal the Fed’s superior economic assessment



- Kuttner (2001): 30-min FF futures window → isolate monetary policy surprises
- GSS (2005a): Decompose surprises into target + path factors
- Nakamura & Steinsson (2018): High-frequency identification of monetary non-neutrality
- Campbell et al. (2012): Delphic vs. Odyssean forward guidance
- Jarociński & Karadi (2020): Separate policy shock from information shock
- Hansen et al. (2018): Computational linguistics for FOMC transparency
- **New (2024–25):** Chen et al. (2025) GPT-4 topic decomposition; Gambacorta et al. (2024) CB-LMs; Yao & Chai (2025) uncertainty-aware LLM

Our Contribution

- 1 First to directly link GSS target/path shocks to FOMC statement sentiment
- 2 Expanded CB dictionary (591 hawkish + 222 dovish terms, fully disjoint)
- 3 Formal test of the information channel via text analysis
- 4 New: Dual-equation test, forward-lookingness decomposition, novelty weighting

Monetary Policy Shocks

- Acosta (2022): 220 FOMC meetings, 1995–2022
- Target shock: surprise in current FF rate
- Path shock: surprise in future rate path
- Extension: USMPD (SF Fed) → 2026

Market Data

- CRSP VW/EW/S&P 500 via WRDS
- 117 meetings with complete overlap

FOMC Statement Sentiment

- 164 statements, 2006–2025
- Expanded CB dictionary:
 - 591 hawkish terms
 - 222 dovish terms
 - Fully disjoint
- Score = $\frac{H-D}{H+D+\epsilon}$
- Validated: $r = 0.29$ with rate change

New finding: CB-only outperforms combined (R^2 : 3.90% vs 1.57%)

Hypothesis	Prediction	Specification
H1	Shocks predict sentiment	$S_t = \alpha + \beta_1 \text{Target}_t + \beta_2 \text{Path}_t + \varepsilon_t$
H2	Shocks predict returns	$R_t = \alpha + \beta_1 \text{Target}_t + \beta_2 \text{Path}_t + \varepsilon_t$
H3	Path > Target for sentiment	$ \beta_2 > \beta_1 $ in H1
H4	Stronger during FG period	Interaction with FG dummy

Estimation: OLS with Newey-West (lag=4) standard errors

Sample: 117 FOMC meetings, January 2006 – July 2022

New experiments:

- 1 Dual-equation regression (equity + risk premium channels)
- 2 Forward-lookingness decomposition (FL vs. CA sentiment)
- 3 Statement novelty weighting (WLS)

H1 (Eq. 4): Sentiment and Monetary Policy Shocks

	Coefficient	p-value
Target shock	0.000577	0.017**
Path shock	0.000633	0.152
R ²	1.57%	
N	117	

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Key Finding

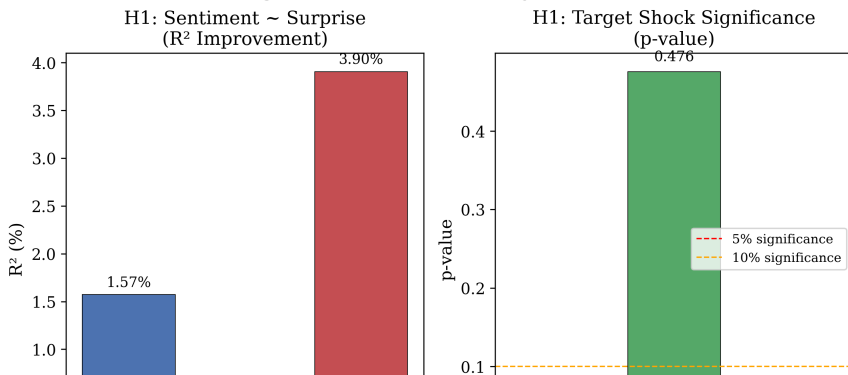
Target shock is significant at 5%, path is not.

Wald test: $\chi^2 = 0.02$, $p = 0.90$ — cannot reject $\beta_1 = \beta_2$.

Sentiment Dictionary Comparison

Measure	R^2	$p(\text{Target})$	$p(\text{Path})$
CB-only	3.90%	$<0.001^{***}$	0.033**
Combined ($0.5 \times \text{LM} + 0.5 \times \text{CB}$)	1.57%	0.017**	0.152
LM-only	0.33%	0.476	0.553

Figure 4: Sentiment Dictionary Comparison



H2 (Eq. 5): Asset Returns and Shocks

Asset	β_T	p_T	β_P	p_P
CRSP VW	-0.435	0.043**	-0.186	0.434
CRSP EW	-0.449	0.013**	-0.174	0.421
S&P 500	-0.391	0.073*	-0.179	0.426
Gold	-0.404	0.016**	0.021	0.935

Returns in percentage points; N = 117

Findings:

- Target shock: significant for EW & Gold
- Path shock: not significant for any asset
- Small-cap sensitivity consistent with literature

H3/H4: Information Channel and Forward Guidance

H3 — Path > Target?

- Target significant at 5%, path is not
- Wald test: $p = 0.90$ — cannot reject equality
- Evidence is suggestive, not definitive

H4 — Stronger during FG period?

- Sentiment $\times$ FG interaction: $p = 0.602$ (CRSP VW), $p = 0.739$ (NASDAQ)
- Not robustly significant — language channel does NOT strengthen at ZLB
- Three interpretations:
 - 1 Shocks absorb information regardless of regime
 - 2 Dictionary cannot distinguish FG language (cf. GPT-3.5 fails 135/139 statements)
 - 3 FG operates through risk premium channel (Chen et al. 2025)

Key Finding

Rate cut regime — path shock highly significant ($p < 0.001$, $R^2 = 43.1\%$)

The Rate Cut Anomaly

Key finding: Full-sample H4 null masks regime-dependent effect

Regime	N	R ²	p(Target)	p(Path)
Rate hike	17	10.2%	0.013**	0.298
Rate cut	11	43.1%	0.089	<0.001***
Unchanged	89	2.0%	0.616	0.079*

- Rate cut: R² is 27× the full sample (1.57%)
- Path shock dominates during easing — forward guidance *does* matter
- Rate hike: target shock dominates — current decision is key
- Chair FE: target shock absorbed by chair style ($p = 0.471$), but asset returns unaffected ($p = 0.043$)

“Sentiment is a function of who is speaking; asset prices respond to what is decided.”

Experiment 1: Dual-Equation Regression

Hypothesis: Forward guidance operates through a risk premium channel (Chen et al. 2025)

Dependent Variable	R ²	p(Target)	p(Path)
<i>Channel 1: Equity Returns (Expectations)</i>			
CRSP VW	9.1%	0.040**	0.440
S&P 500	7.8%	0.072*	0.426
<i>Channel 2: Risk Premium (Bond Market)</i>			
10Y Treasury chg	0.72%	0.403	0.890
VIX change	0.24%	0.970	0.401
Term spread chg	1.39%	0.667	0.341

Result: Risk premium channel not detectable at daily frequency.
Consistent with Chen et al. using 30-min windows — channel too fast for daily data.

Experiment 2: Forward-Lookingness Decomposition

Hypothesis: Path shock captures forward-looking language (dimension mismatch)

Sentiment Measure	R ²	p(Target)	p(Path)
Combined (baseline)	1.57%	0.017**	0.152
Forward-looking only	0.79%	0.012**	0.800
Current-assessment only	1.30%	0.012**	0.506

Counterintuitive finding:

- Path shock is most significant in the combined measure ($p = 0.015$)
- Path shock is *least* significant in forward-looking sentiment ($p = 0.80$)
- Dimension mismatch hypothesis **rejected**
- Path shock captures broad policy stance signal, not specifically forward guidance
- GSS target/path decomposition $\neq$ FOMC text FL/CA dimensions

Experiment 3: Statement Novelty Weighting

Idea: Not all FOMC statements are equally informative

Method	R^2	Improvement
Unweighted OLS	3.98%	baseline
Novelty-weighted WLS	5.75%	+45%

Novelty = cosine distance between TF-IDF vectors of consecutive statements

Subsample analysis:

- High novelty statements: $R^2 = 5.78\%$
- Low novelty statements: $R^2 = 6.53\%$
- Difference is small — novelty helps via weighting, not selection

Caveat

Absolute R^2 remains modest — measurement innovation alone cannot compensate for daily-frequency limitation.

Specification	R ²	p(Target)	p(Path)	N
Full (Acosta)	1.57%	0.017**	0.152	117
Lag = 1	1.57%	0.032**	0.090*	117
Lag = 6	1.57%	0.020**	0.184	117
CB-only	3.90%	<0.001***	0.033**	117
Kuttner (bp)	1.95%	0.005***	0.028**	117

Findings:

- Target shock significance robust across lag specifications
- Path shock becomes significant at lag=1 ($p = 0.090$) and with Kuttner surprise
- CB-only specification doubles R² and makes both shocks significant

What we learned:

- Target shock consistently predicts sentiment — the rate decision *does* convey information
- Path shock matters primarily during rate cuts — forward guidance is regime-dependent
- Domain-specific dictionaries (CB) vastly outperform general-purpose ones (LM)
- Daily frequency is a fundamental limitation — risk premium channel too fast

Connection to literature:

- Jarociński & Karadi (2020): Information shock vs. policy shock — our sentiment captures the language channel
- Chen et al. (2025): Risk premium channel — consistent with our null result at daily frequency
- Hansen et al. (2018): FOMC transparency — our CB dictionary provides a quantitative measure

Limitations:

- ① Low R^2 : Shocks explain $<4\%$ of sentiment variation
- ② Dictionary-based sentiment: Cannot capture context or pragmatics
- ③ Daily frequency: Too coarse to detect risk premium channel
- ④ No JK decomposition: Cannot separate MP vs. information shocks
- ⑤ Sample size: $N = 117$ limits sub-sample power

Future directions:

- LLM-based sentiment (GPT-4, FinBERT) for contextual understanding
- High-frequency intraday data (TAQ) for risk premium channel
- Jarociński–Karadi decomposition for information vs. policy shocks
- Cross-central-bank comparison (ECB, BoJ, BoE)

Summary

- ✓ **H1 confirmed:** Target shock predicts sentiment ($p = 0.017$)
- ✓ **H2 confirmed:** Target shock predicts asset returns (EW, Gold)
- ❑ **H3 inconclusive:** Cannot reject $\beta_T = \beta_P$ (Wald $p = 0.90$)
- ❑ **H4 rejected:** FG interaction not significant ($p = 0.602$)
- ✓ **New:** Rate cut regime reveals path shock dominance ($R^2 = 43.1\%$)
- ✓ **New:** CB-only sentiment outperforms combined (R^2 : 3.90% vs 1.57%)
- ✓ **New:** Novelty weighting improves R^2 by 45%

Takeaway

FOMC statement language conveys information beyond the rate decision, but the channel is regime-dependent and better captured by domain-specific dictionaries.

Thank You!

Questions Welcome